

Functional forms and interactions

Slide route: Lecture7v2.tex: polynomials, logarithms, and interactions through wrapping up. Chapter 8.

Session 9 | A slope need not be constant

Outcomes: Interpret quadratic terms, distinguish log specifications, and calculate a conditional effect in an interaction model.

100-minute route

0–10	Retrieve coefficient units and joint tests.
10–30	Nonlinear functions and polynomials.
30–50	Linear-log, log-linear, log-log examples.
50–55	Pause.
55–80	Dummy/continuous and continuous/continuous interactions.
80–90	Integrate interpretation with tests.
90–100	Exit exercise and Midterm 2 preparation.

Slide-by-slide script

Slide 1 | Title

Guide Open with a picture of a curved relationship and ask why one constant slope can miss it.

Slide 2 | Roadmap

Guide Preview polynomials, logarithms, and interactions. Tell students every interpretation must name the relevant value of X or Z .

Slide 3 | Last lesson

Guide Retrieve a partial coefficient and a joint restriction. Explain that functional form changes the question the coefficient answers.

Slide 4 | Last lesson

Guide Use the OVB sign rule to motivate why a flexible model is not a cure for confounding.

Slide 5 | Introduction

Guide Distinguish linear in parameters from linear in the regressor. This class uses OLS with nonlinear transformations.

Slide 6 | Linear fit on a linear relationship

Guide Show a constant marginal effect and ask where it appears in the equation.

Slide 7 | Linear fit on a nonlinear relationship

Guide Use a curved data-generating relationship to show why residual patterns can reveal misspecification.

Slide 8 | General nonlinear population function

Guide Introduce $\mathbb{E}(Y \mid X = x)$ as the object a flexible conditional mean describes.

Slide 9 | Nonlinear functions of one variable

Guide Ask students to calculate a derivative before interpreting a coefficient.

Slide 10 | Polynomials

Guide Write $10 + 2x - .05x^2$ and calculate the marginal effect at $x = 10$.

Slide 11 | Polynomials

Guide Find the turning point $x = 20$ and ask whether it lies inside the observed support.

Slide 12 | Polynomials

Guide Calculate the exact change from 10 to 11, .95, and compare it with the derivative at 10.

Slide 13 | Polynomials

Guide Explain that including x and x^2 is still linear regression in the coefficients.

Slide 14 | Log functions

Guide State which variable is logged before interpreting a coefficient. Never say “one percent” without checking the specification.

Slide 15 | Log functions

Guide For a linear-log model, a one-unit change in X is associated with approximately $100\beta_1/Y$ percent when Y is in levels; use the deck’s exact convention.

Slide 16 | Log functions

Guide For a log-linear model, a one-unit change in X changes Y by approximately $100\beta_1$ percent.

Slide 17 | Log functions

Guide For a log-log model, interpret β_1 as an elasticity: a 1 percent change in X is associated with about β_1 percent in Y .

Slide 18 | Log-outcome example

Guide Use .10 to calculate the exact change $100(e^{.10} - 1) \simeq 10.52\%$ and compare it with the approximation.

Slide 19 | Log-outcome example

Guide Ask students to identify the units of the coefficient and the reference change in X .

Slide 20 | Log-outcome example

Guide Warn that log interpretations are not causal without a credible identifying design.

Slide 21 | Comparing log specifications

Guide Have students translate each equation into a sentence before comparing the coefficients.

Slide 22 | Interactions

Guide Explain why the effect of one variable can depend on another.

Slide 23 | Two dummies

Guide Write the four predicted group means and identify the omitted reference group.

Slide 24 | Two dummies

Guide Use the interaction coefficient as a difference-in-differences contrast in the descriptive model.

Slide 25 | Two dummies

Guide Ask whether the interaction coefficient is a causal DiD effect; answer only with the required design assumptions.

Slide 26 | Dummy and continuous variable

Guide For $Y = \beta_0 + \beta_1 X + \beta_2 D + \beta_3 XD$, calculate the group contrast at $X = 4$.

Slide 27 | Dummy and continuous variable

Guide Use the example $3 + 4x$ for the program effect and explain why 3 is only the effect at $x = 0$.

Slide 28 | Dummy and continuous variable

Guide Calculate the slope for each group: β_1 and $\beta_1 + \beta_3$.

Slide 29 | Dummy and continuous variable

Guide Test the effect at $x = 4$ as a linear restriction, not by testing only β_2 .

Slide 30 | Dummy and continuous variable

Guide Ask how centering X changes the interpretation of the main dummy coefficient.

Slide 31 | Dummy and continuous variable

Guide Show that centering changes parameterization but not fitted values.

Slide 32 | Continuous interactions

Guide Interpret $\partial \mathbb{E}(Y | X, Z) / \partial X = \beta_1 + \beta_3 Z$.

Slide 33 | Continuous interactions

Guide Give a numerical example with $Z = 2$ and $Z = 5$ so students see the slope move.

Slide 34 | Continuous interactions

Guide Remind students to report the value of the interacting variable with every marginal effect.

Slide 35 | Continuous interactions

Guide Discuss extrapolation: do not interpret effects far outside the observed support.

Slide 36 | Continuous interactions

Guide Connect conditional effects to joint hypothesis tests for several coefficients.

Slide 37 | Wrapping up

Guide Summarize polynomial, log, and interaction interpretations in a three-row board table.

Slide 38 | Wrapping up

Guide Ask which models remain linear in parameters and can be estimated by OLS.

Slide 39 | Wrapping up

Guide Use one final translation exercise and insist on units.

Slide 40 | Wrapping up

Guide Tie the functional form back to fit, prediction, and causal interpretation.

Slide 41 | Linear versus non-linear models

Guide Close with the distinction between a nonlinear conditional mean and a nonlinear estimator.

Slide 42 | Importance of interactions

Guide Have students state one setting where a treatment effect should plausibly vary across people.

Worked example | Quadratic experience

Use with slides Lecture7v2 slides 8–13: Nonlinear Functions and Polynomials in X . Use the derivative and turning-point calculation on slides 9–13; pause before moving to logs.

For the synthetic fitted relationship

$$\widehat{wage} = 10 + 2x - .05x^2, \quad \frac{\partial \widehat{wage}}{\partial x} = 2 - .1x,$$

the marginal effect is 1 at $x = 10$ and 0 at $x = 20$. The turning point is 20. The exact fitted change from 10 to 11 is

$$2 - .05(11^2 - 10^2) = .95,$$

not exactly the derivative at 10. Only discuss the turning point substantively if it lies within a relevant supported range.

Board comment The model is nonlinear in x but linear in the coefficients. OLS can estimate it after including both x and x^2 .

Log interpretation crib

Model	Interpretation of β
$Y = \alpha + \beta \log X + u$	1% higher X : approximately $.01\beta$ units higher Y .
$\log Y = \alpha + \beta X + u$	One unit higher X : approximately $100\beta\%$ higher Y .
$\log Y = \alpha + \beta \log X + u$	β is the elasticity of the modeled log relationship.

Worked examples | Logs and heterogeneous effects

Use with slides Lecture7v2 slides 14–21 for the log examples and slides 22–39 for interactions. Use the log-outcome calculation on slides 18–21, then use the program-by-experience example on slides 26–38.

If a log-outcome model has coefficient .08, a one-unit increase changes the fitted log index by .08. The corresponding multiplicative change is

$$100(e^{.08} - 1) \simeq 8.33\%,$$

versus the 8% approximation. Exponentiating a fitted conditional log mean does not automatically give the conditional arithmetic mean of Y ; retransformation requires attention to the error distribution.

For a level-log model with coefficient 20, a 10% increase in X implies $20 \log(1.10) \simeq 1.906$ outcome units, compared with the approximation 2.

Interaction board example

Let D be a program indicator and x be experience:

$$\hat{Y} = 10 + 2x + 3D + 1.5(xD).$$

For $D = 0$, the line is $10 + 2x$; for $D = 1$, it is $13 + 3.5x$. The program-group contrast at experience x is $3 + 1.5x$, so it equals 9 at $x = 4$.

Ask Is 3 the program effect for everyone? No, it is the contrast at $x = 0$. Is 1.5 the entire program effect? No, it is the difference in slopes. A causal reading still needs identification assumptions.

Test at $x = 4$ Test $\beta_D + 4\beta_{xD} = 0$, using the covariance of the two estimates. Testing only the main dummy coefficient does not answer this question.

Continuous interaction For $Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + u$, the X derivative is $\beta_1 + \beta_3 Z$. State the value of Z whenever reporting it.

Exit answers A log-log slope of $-.7$ implies roughly a .7% decrease for a 1% increase in X . A quadratic model can be estimated by OLS. An interaction coefficient is not a universal total effect.

If short Use one example of each functional form and one interaction in depth; assign repetitive interaction plots for self-study. Proposed Midterm 2 coverage is Lectures 5–7, using prerequisites from Lectures 1–4.

Source: Lecture7v2.tex; synthetic numerical illustrations.